

Program9

Organize kubernetes pods using labels and namespace

Project Structure:

```
-----program9
      ----- namespace.yaml
      ----- pod.yaml
```

```
Dec 19 10:34 AM
isha@meow: ~/DevOps/prg9
isha@meow:~/DevOps/prg9$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
👤 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner
👏 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step1: create a folder

Step2: create a namespace.yaml and pod.yaml

```
>> nano namespace.yaml
```

```
>> nano pod.yaml
```

Step3: Apply namespace and pod.yaml file

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program9
GNU nano 7.2 namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step4: verify pods by namespace

```
GNU nano 7.2 pods.yaml *
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
```

Help Write Out Where Is Cut Execute Location Undo Set Mark
Exit Read File Replace Paste Justify Go To Line Redo Copy

Step5: Apply namespace and pod.yaml file

Step6: Verify the pod by namespace

```
isha@meow:~/DevOps/prg9$ nano namespaces.yaml
isha@meow:~/DevOps/prg9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
isha@meow:~/DevOps/prg9$ kubectl get namespaces
NAME                STATUS    AGE
default             Active   17d
dev                 Active   7s
kube-node-lease     Active   17d
kube-public         Active   17d
kube-system         Active   17d
prod               Active   7s
isha@meow:~/DevOps/prg9$ nano pods.yaml
isha@meow:~/DevOps/prg9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
isha@meow:~/DevOps/prg9$ kubectl get pods -n dev
NAME                READY    STATUS    RESTARTS   AGE
backend-pod         0/1      ContainerCreating    0           7s
frontend-pod        0/1      ContainerCreating    0           7s
isha@meow:~/DevOps/prg9$ kubectl get pods -n prod
NAME                READY    STATUS    RESTARTS   AGE
backend-pod         1/1      Running    0          18s
frontend-pod        1/1      Running    0          18s
```

Step7: Check the status

```
pod/backend-pod created
isha@meow:~/DevOps/prg9$ kubectl get pods -n dev
NAME          READY   STATUS             RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0           7s
frontend-pod  0/1     ContainerCreating   0           7s
isha@meow:~/DevOps/prg9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           18s
frontend-pod  1/1     Running   0           18s
```

Step8: Filter pod by labels

```
isha@meow:~/DevOps/prg9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           37s
isha@meow:~/DevOps/prg9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           43s
isha@meow:~/DevOps/prg9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           70s
frontend-pod  1/1     Running   0           70s
isha@meow:~/DevOps/prg9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           79s
frontend-pod  1/1     Running   0           79s
isha@meow:~/DevOps/prg9$
```

Step9: Clear all the process and delete all pods and services.